AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 23 · STUDENT LESSON

Recommendation Scores

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Check eligibility. Calculate components. Explain the choice.

Rank only permitted activities under explicit preference weights, tie rules and empty-result handling.

Keep one habit: Opening constraint → separate weight-sensitive tie → full audit → independent fresh limit.

MAIN LESSON · SECTION 1

Which choices qualify before their scores are compared?

> Keep mandatory requirements separate from weighted preferences.

Your learning goals

I can compute a **weight**ed **score**.

I can apply hard **constraint**s before **rank**ing.

I can explain how **weight** choices change outcomes.

Recall the direction of a distance comparison

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a **score** for your ability.

- Which is closer under absolute-gap distance: a gap of 2 or a gap of 5?

Your end goal

By the end, I can **rank** only eligible options with a transparent **weight**ed **score**.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 23: Recommendation **Score**s

A **weight**ed **score** expresses priorities but also embeds choices.
</details>



THINKER MISSION

Create two user profiles and calculate book **rank**ings.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

State the activity score and20-minute requirement

> A preference score cannot compensate for an exceeded time limit.

Feature meanings, scales, formula and hard limit

A fictional activity chooser uses enjoyment and learning ratings from 0 to 5. Its **score** is 2 times enjoyment plus learning. The activity must also take at most 20 minutes.

Duration is a hard requirement; a high **score** cannot compensate for exceeding it.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use recommendation **score**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

MAIN LESSON · SECTION 3

Trace eligibility before the score calculation

> Only the eligible activity is printed by this minimal example.

Predict and run the complete opening script

```
options = [{"name": "A", "fun": 4, "learning": 2, "minutes": 15},
           {"name": "B", "fun": 3, "learning": 5, "minutes": 25}]
for item in options:
    if item["minutes"] <= 20:
        score = 2 * item["fun"] + item["learning"]
        print(item["name"], score)
```

Explain the inclusive boundary and ordered decisions

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
A	2 * 4 + 2 = 10	Eligible at 15 minutes
B	2 * 3 + 5 = 11	Excluded at 25 minutes

Read each row from left to right and connect the evidence to the stated rule.

Only A is printed, with **score** 10. If B took 20 minutes it would qualify and have the higher **score** 11. The inclusive boundary matters.

Ratings and **weight**s are invented for practice and require justification before use in a real decision.

Make two decisions in order. First check the duration: A takes 15 minutes and qualifies under the 20-minute limit; B takes 25 minutes and does not. Then calculate **score**s for eligible choices. A **score**s 2 times 4 plus 2, or 10.

B would **score** 11, but that **score** cannot remove the time requirement. Keep the rejection reason beside B instead of calling it a worse-scoring activity.

State input assumptions and omitted review handling

This short script assumes the named fields are present, fun/learning are valid whole-number0–5 ratings and minutes is a nonnegative whole number. It demonstrates the time constraint and scoring; it is not a complete input validator. A missing field must not be silently treated as0. Keep rejected items and reasons in the investigation record even though this minimal script prints only eligible choices.

Explain a deciding step

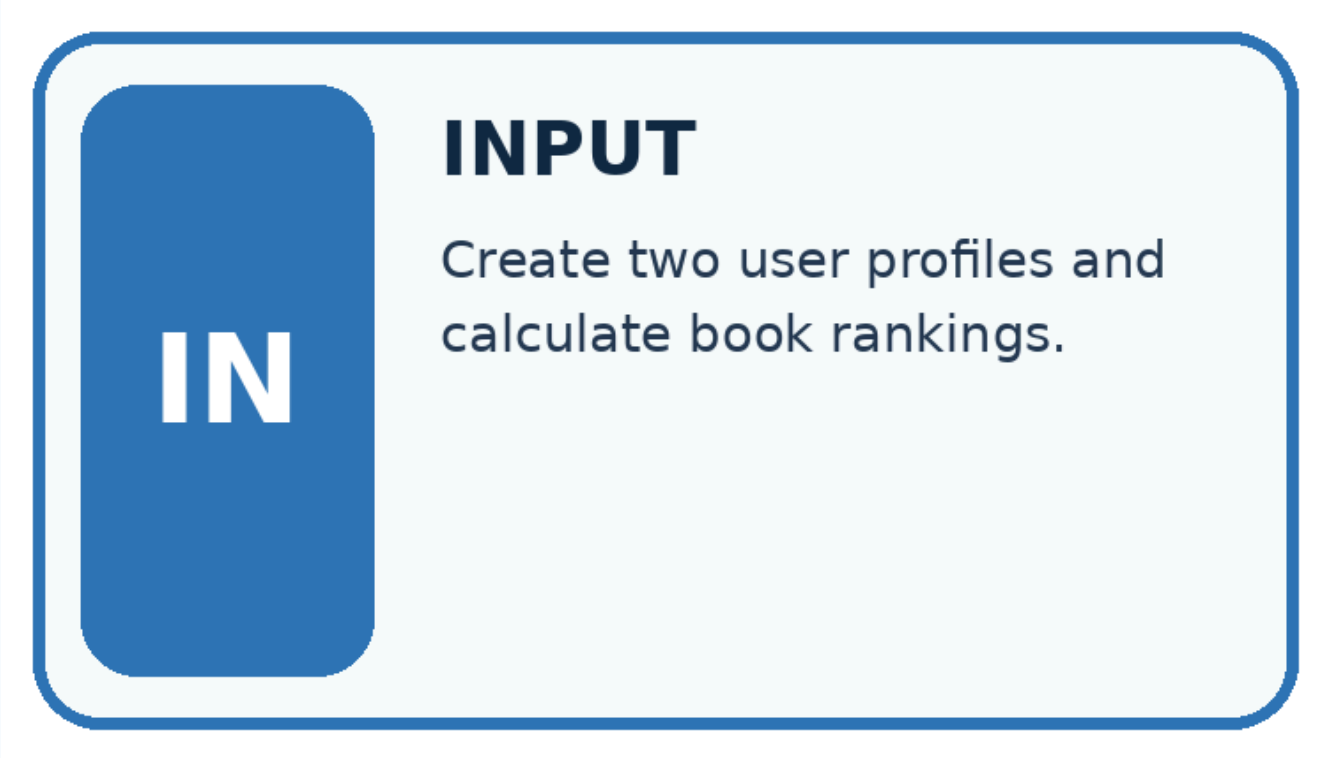
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → **Process** → **Output**

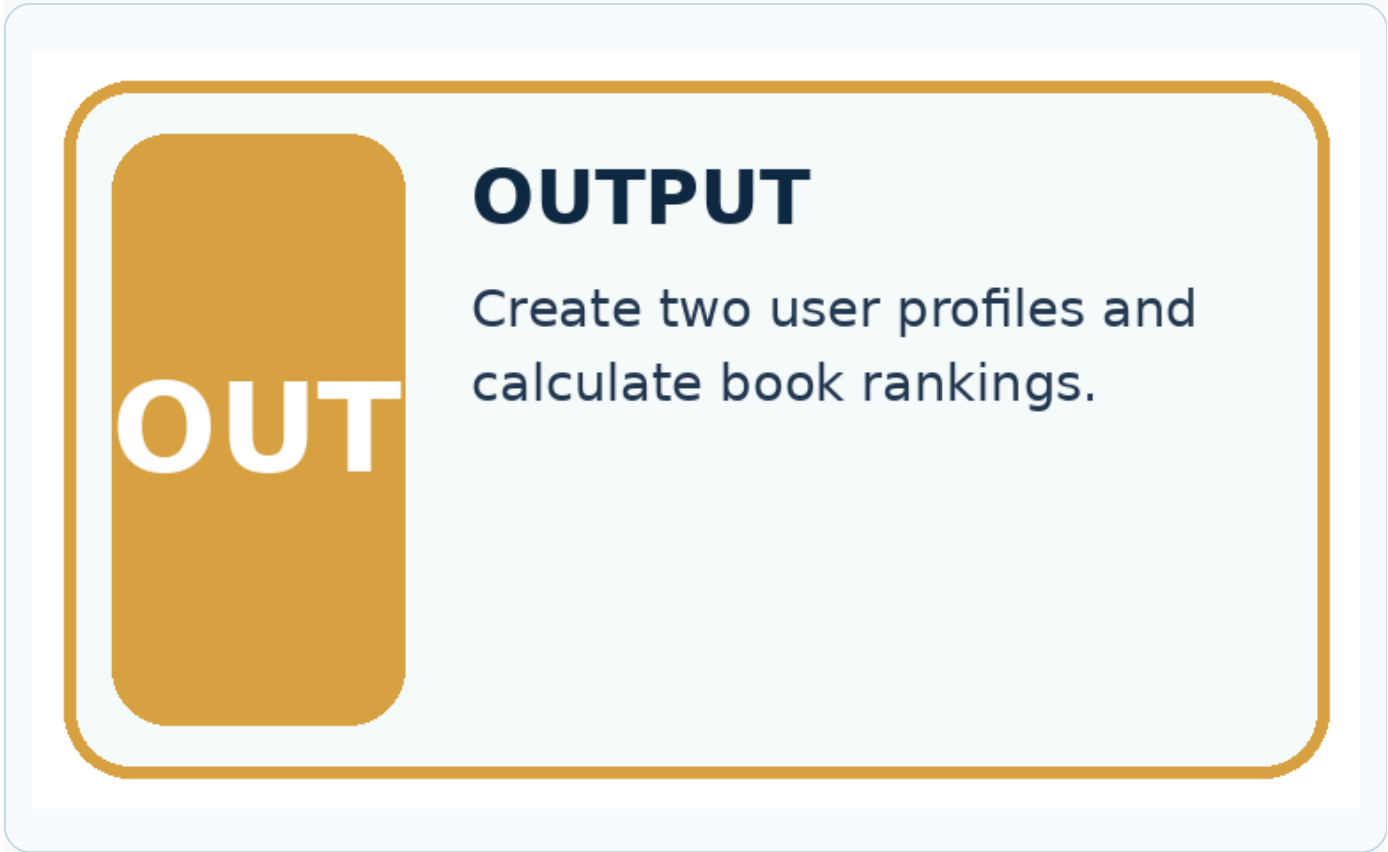
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
A weighted score expresses priorities but also embeds choices.

↓


OUT
OUTPUT
Create two user profiles and calculate book rankings.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Check must-have rules before scoring

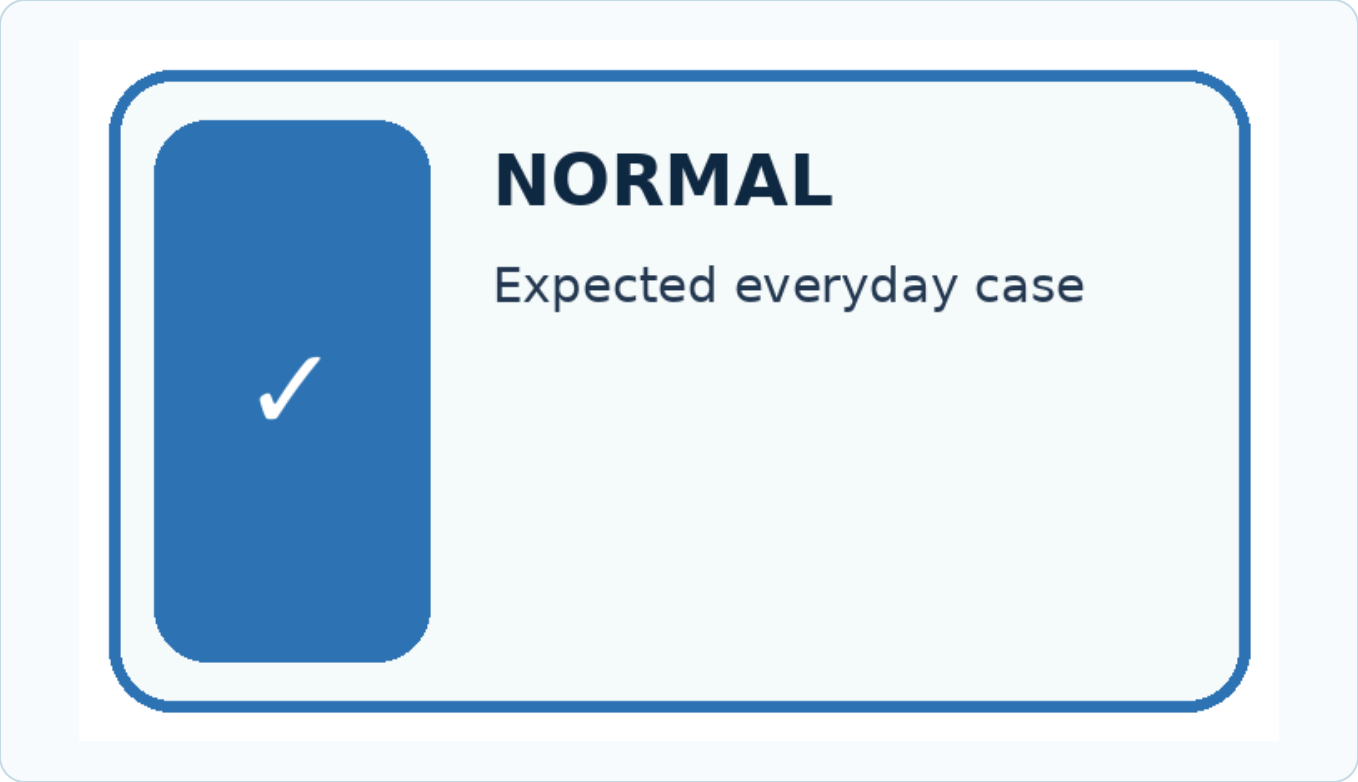
Connect the picture and the explanation

MAIN LESSON · SECTION 4

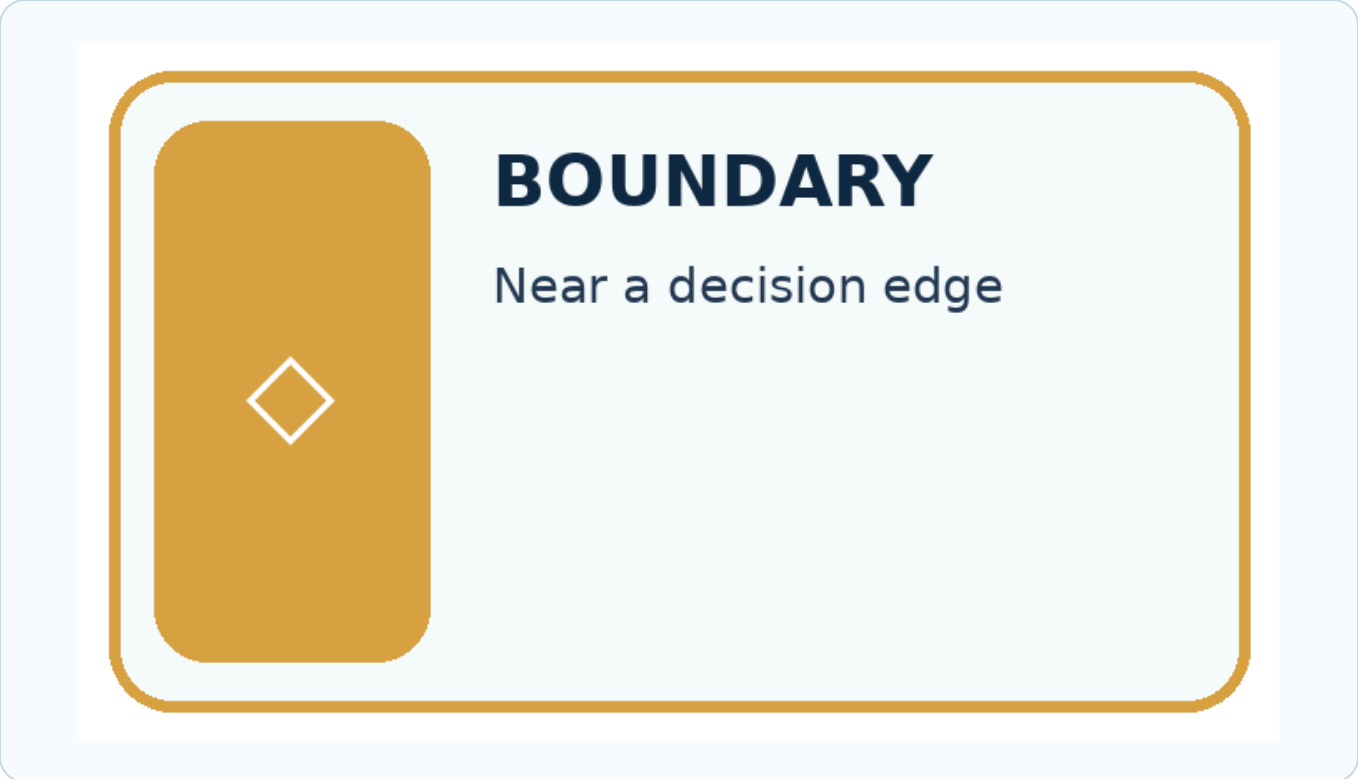
Challenge an excluded high score and an exact boundary

> Changing a preference does not silently change a hard requirement.

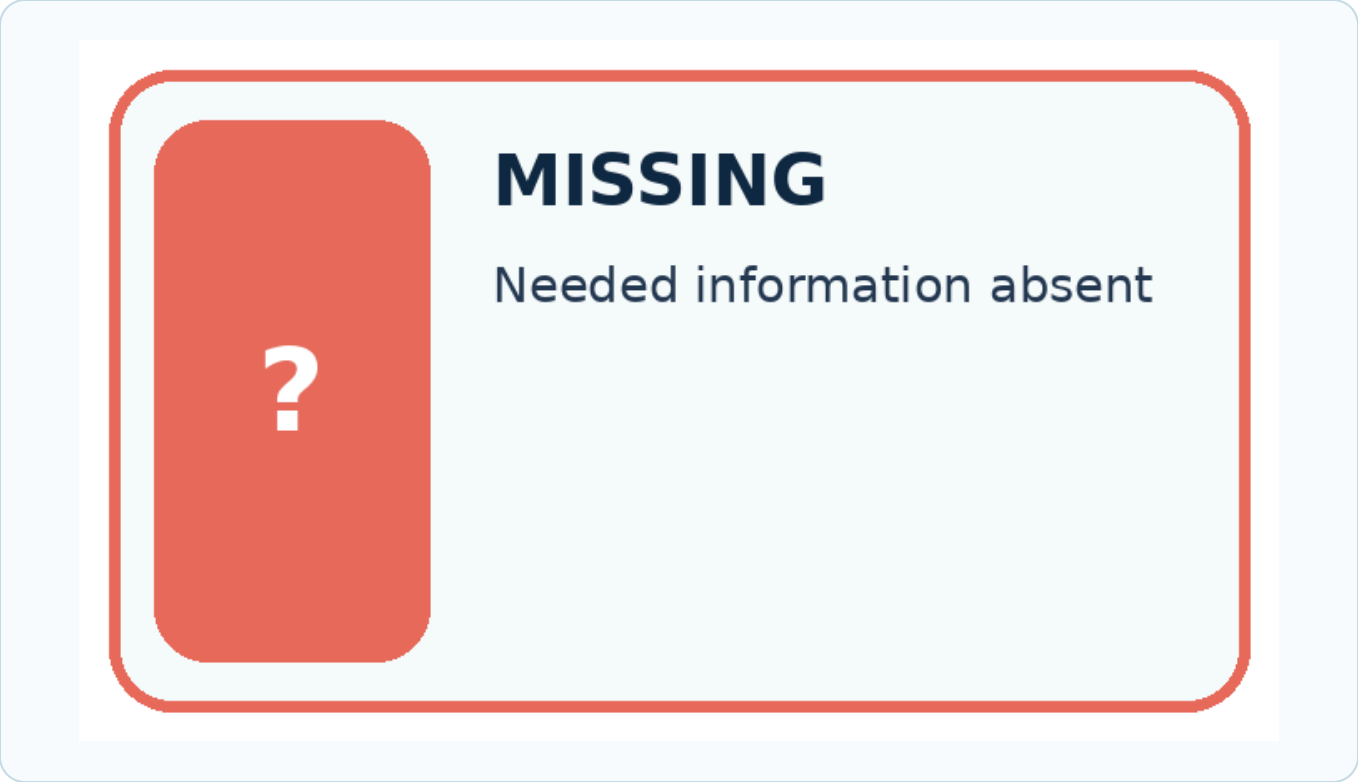
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Change interest **weight** from 2 to 3, keeping time limit 14.

▼ Check after predicting

Prediction: A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

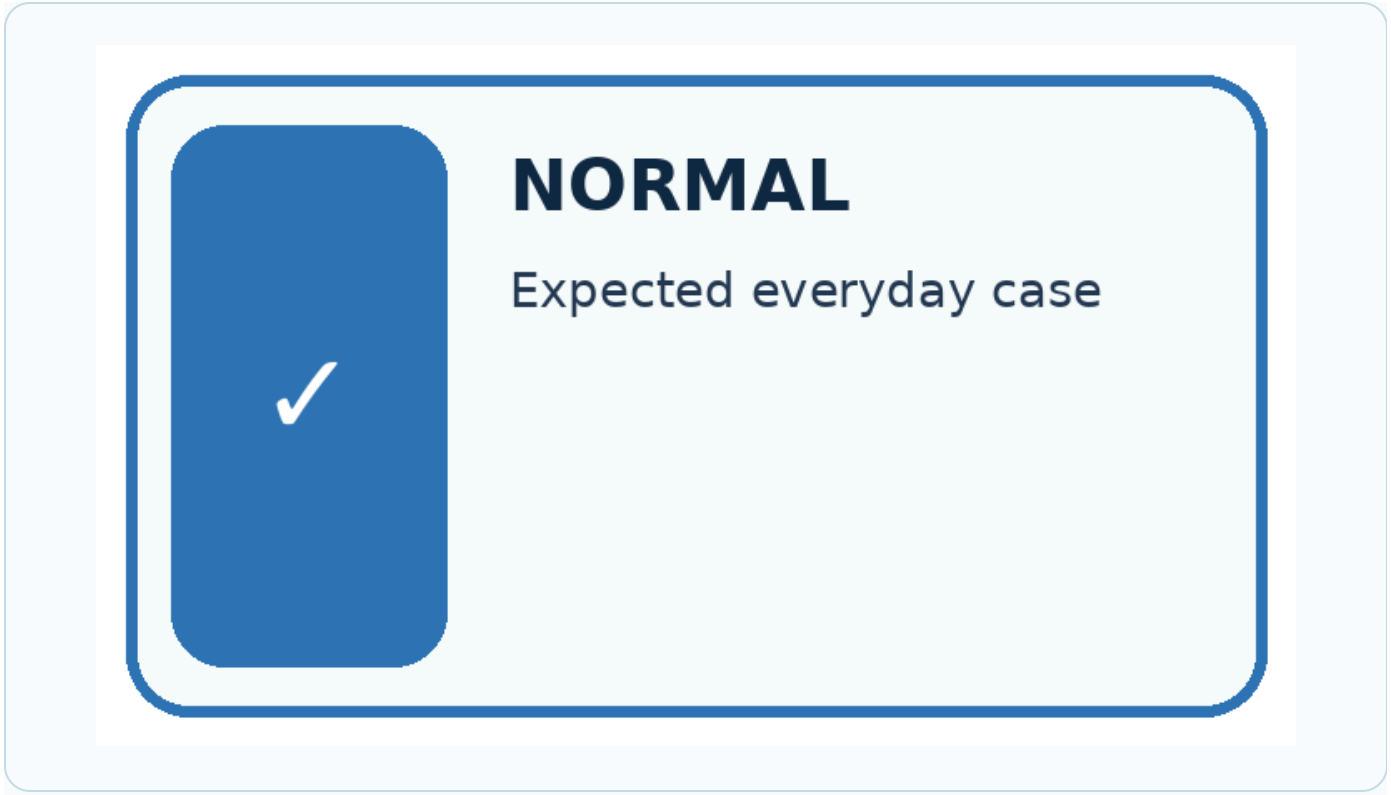
Reason: B remains too long; changing preferences changes only **rank**ing of A and C.

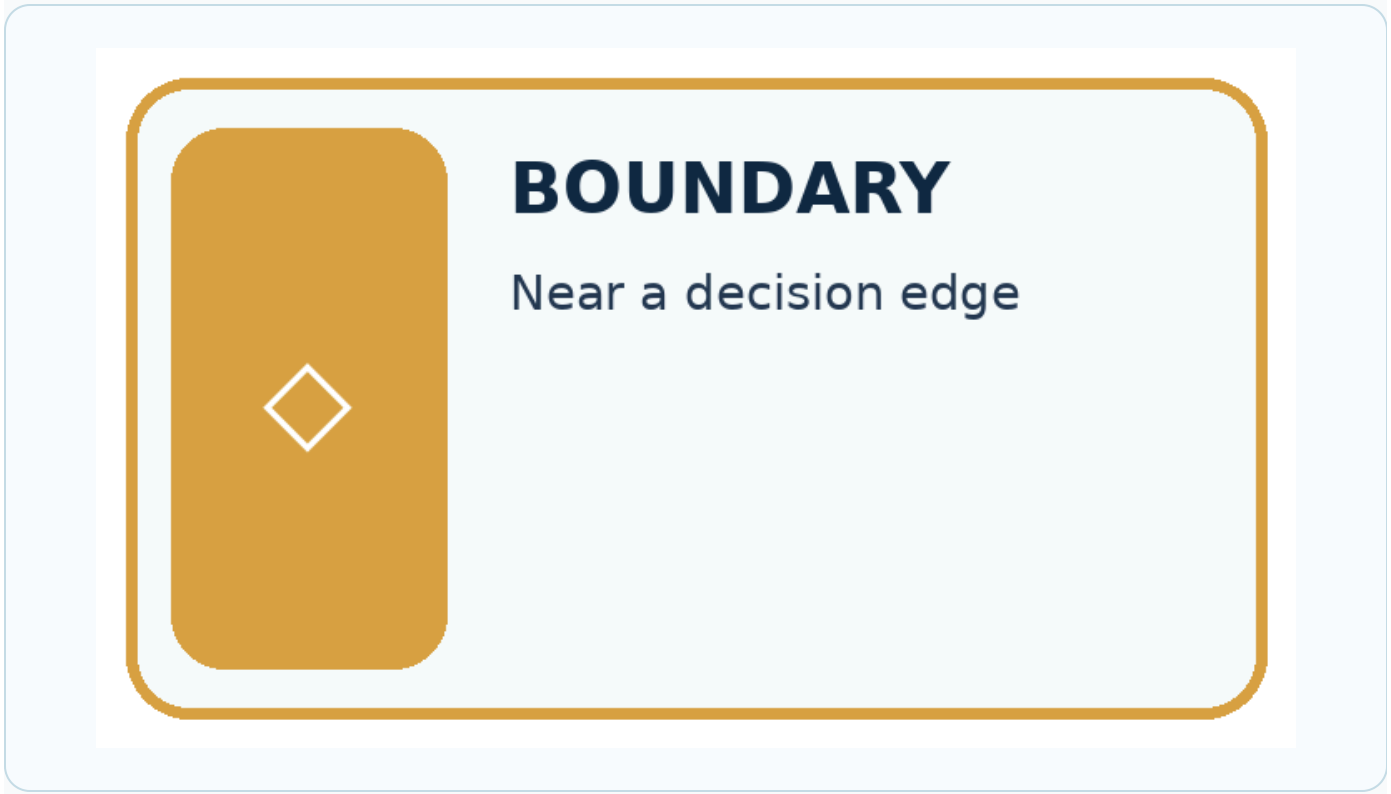
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

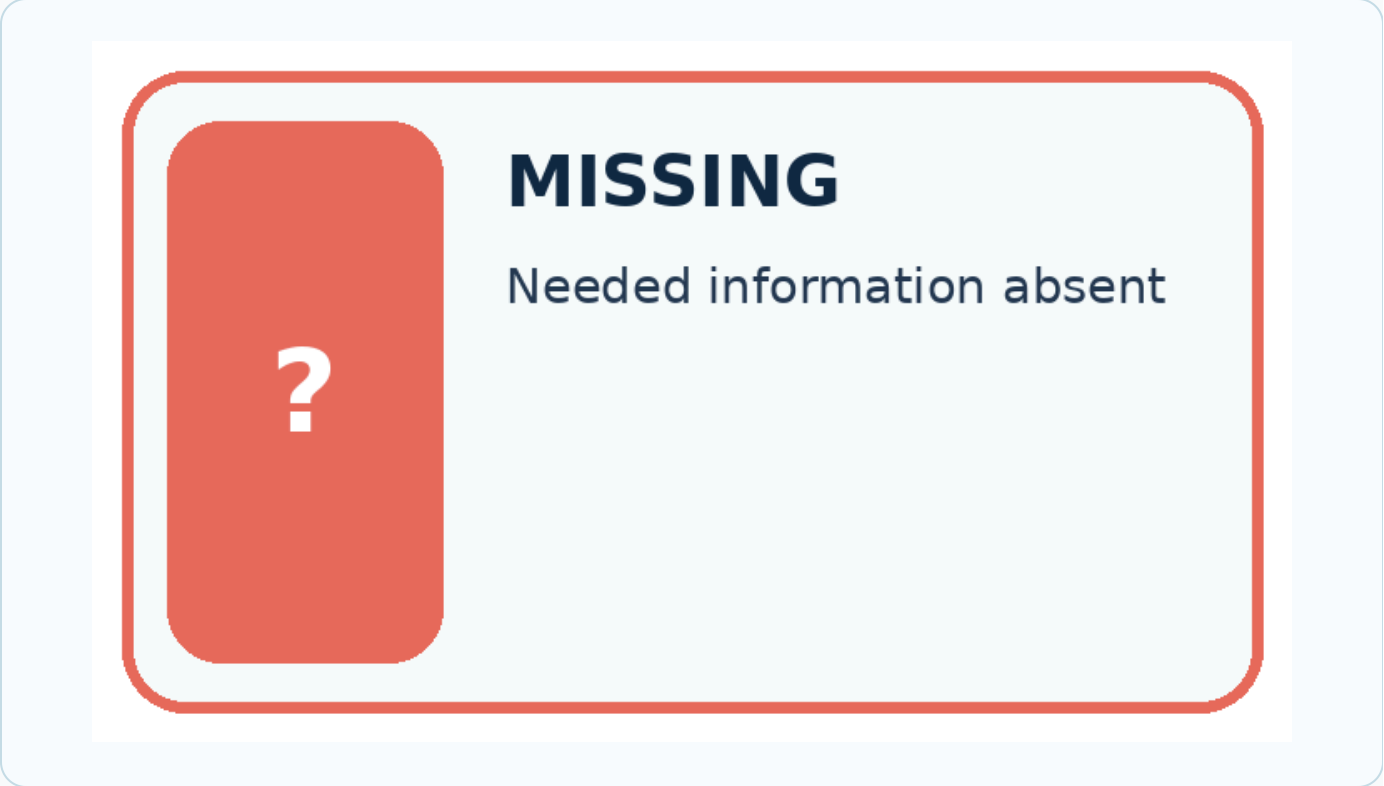
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Change weights while keeping eligibility visible

> Trace the separate pictured interest/practice comparison.

Try the opening weight change before discussing

> **If enjoyment weight becomes 1 and learning weight becomes 2, find both scores before applying the same duration limit.**

Separate the different activity records

Keep the cases distinct: opening activities use fun/learning with a20-minute limit and weights(2,1); the separate pictured choices below use interest/practice and a14-minute limit. Independent W4 later uses its own15-minute limit and ratings. NamesA/B/C do not mean the same rows in every case.

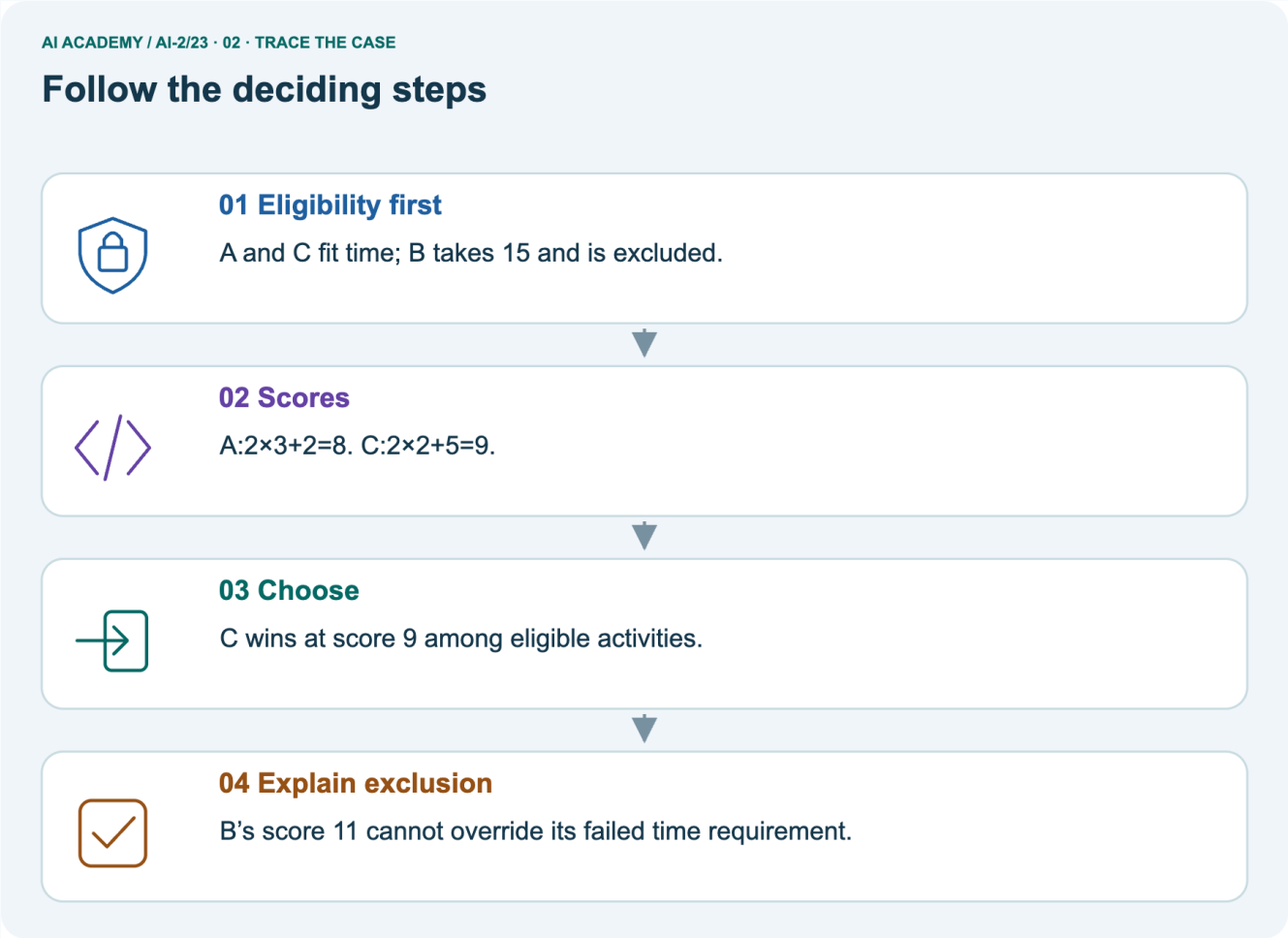
Read the14-minute pictured contract

A fictional book-activity chooser has a 14-minute limit.

Score=2×interest+practice. A=(interest3,practice2,time14); B=(4,3,15); C=(2,5,12).

Require whole nonnegative feature values; time must be at most the limit. Highest **score** wins; tie by ID.

Trace eligible scores and exclusion



Follow the labelled picture with the matching explanation. Open the vector diagram

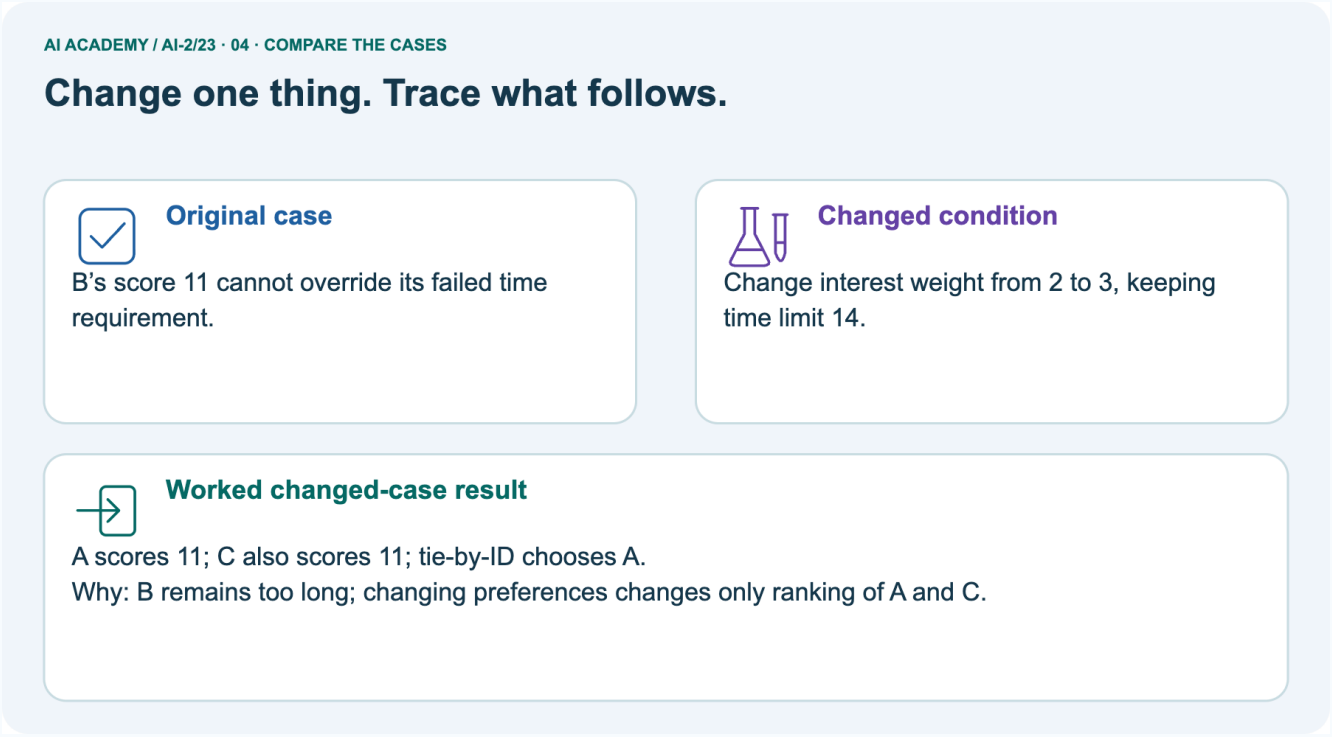
A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

Predict interest weight3 under the same limit

- Change one thing: Change interest **weight** from 2 to 3, keeping time limit 14. Predict the result before checking the explanation. What remains the same?

▼ Check the tie policy after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why B remains excluded and limit the claim

- Read the labelled picture: Why does B not become the winner when its **score** is largest?
- Toy interest **score**s are supplied **input**s, not measured enjoyment or a universal definition of a good activity.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 23 • WORKED EVIDENCE

1. Score = 3×genre match + 2×difficulty match + 1×length match.
2. Book A matches genre and length: 3+0+1=4.
3. Book B matches difficulty and length: 0+2+1=3.
4. A ranks first for these weights—not as a universal best book.

1. **Score** = 3×genre match + 2×difficulty match + 1×length match.

2. Book A matches genre and length: 3+0+1=4.

3. Book B matches difficulty and length: 0+2+1=3.

4. A **rank**s first for these **weight**s—not as a universal best book.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Recommendation Scores

The facts we will use

AI ACADEMY / AI-2/23

Time filtering and scores

	Minutes	Score	Status
A	14	8	eligible
B	15	11	excluded
C	12	9	eligible

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/23

Recommendation Scores

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Eligibility first

A and C fit time; B takes 15 and is excluded.



2. Scores

A:2 x 3+2=8. C:2 x 2+5=9.



3. Choose

C wins at score 9 among eligible activities.



4. Explain exclusion

B's score 11 cannot override its failed time requirement.

CHANGE ONE THING

Change interest weight from 2 to 3, keeping time limit 14.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Eligibility first	A and C fit time; B takes 15 and is excluded.
2. Scores	A:2×3+2=8. C:2×2+5=9.
3. Choose	C wins at score 9 among eligible activities.
4. Explain exclusion	B's score 11 cannot override its failed time requirement.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Weights encode declared preferences, not educational truth

> Explain who chose the rule and retain sensitivity results.

State the preference and benefit limits

Weights represent preferences, not a discovered truth. Explain who chose them and test whether small changes alter the recommendation. Avoid claiming that a larger **score** proves greater educational value.

Calculate the supplied W5 sensitivity comparison

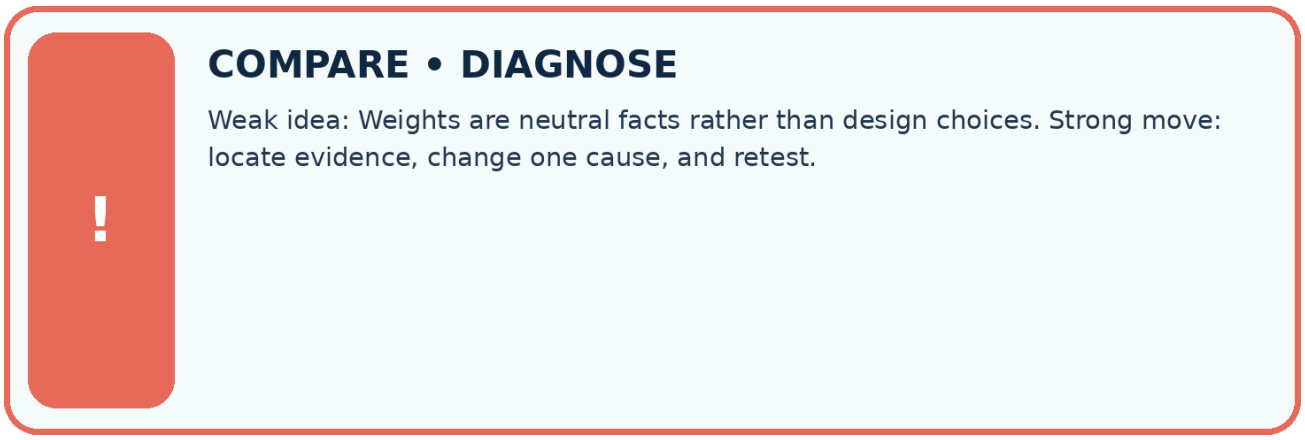
For eligible A (fun 4, learning 1) and B (fun 2, learning 4), compare weights (2,1) with (1,2).

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.



Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 8

Report no eligible option rather than silently relaxing the rule

> A ranking is only among the permitted set.

Check the ordering and empty-result policy

Choosing the highest **score** and only then ignoring its duration breaks the requirement. Filter for eligibility first. If no item qualifies, report no eligible option instead of relaxing a requirement silently.

Practise the standalone duration boundary in W2

Use workbook W2 for the connected example “**Weight**ed **score**s and **constraint**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge a successful-run guarantee

- Use the worked case for **Weight**ed **score**s and **constraint**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured excluded high score

Why does B not become the winner when its **score** is largest?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.

TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Check a boundary

Check your reasoning

Explain the picture

My explanation using a pictured step and the given facts

MAIN LESSON · SECTION 9

Audit two supplied preference settings on the same eligible set

> Keep score components, exclusion reasons, ties and changed versions visible.

The complete input-validation and scoring program

```
options = [{"id": "A", "interest": 3, "practice": 2, "minutes": 14},
           {"id": "B", "interest": 4, "practice": 3, "minutes": 15},
           {"id": "C", "interest": 2, "practice": 5, "minutes": 12}]
limit = 14
for interest_weight in [2, 3]:
    scores = {}
    review = []
    for item in options:
        values = [item.get(key) for key in ["interest", "practice", "minutes"]]
        if not all(type(value) is int and value >= 0 for value in values):
            review.append((item["id"], "missing or invalid input"))
        elif item["minutes"] > limit:
            review.append((item["id"], "exceeds time limit"))
        else:
            scores[item["id"]] = interest_weight * item["interest"] +
item["practice"]
    print("weight", interest_weight, "scores", scores, "review", review)
    if not scores:
        print("No eligible option")
    else:
        highest = max(scores.values())
        tied = sorted(name for name in scores if scores[name] == highest)
        print("top", tied, "chosen", tied[0])
```

Predict, execute, compare weights and test an empty set

1. Use the complete pictured A/B/C records repeated in this script. The interest/practice contract here is whole nonnegative values; IDs are supplied unique text. Predict eligibility and each component before running with interest weights2 and3, practice weight1 and limit14. Keep the data and limit unchanged between these fictional preference settings. all checks that every required value meets the stated input rule before any scoring. max finds the highest eligible score; sorted retains alphabetical tie order. Record all tied top IDs and the disclosed chosen ID, as well as excluded rows and reasons. Then on a separate copy set limit0 to check no eligible option, without treating the changed limit as the original user requirement. No real preference is measured and no model is trained. Reserve the15-minute independent case for W4.

Retain evidence and explain the tradeoff

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → TEST → RECORD → EXPLAIN

Create two reader profiles with different weights, calculate every term, total, rank, and explain why the order changes.

Use the supplied14-minute A/B/C records and complete two-**weight** comparison. Retain eligible scores, excluded rows and reasons, all top ties and the declared ID choice. Test no eligible option on a separately labelled changed-limit copy; do not invent reader measurements.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: use the fresh15-minute constraint and scores

> Calculate the audit scores but choose only among eligible choices.

The independent records and rule

Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate **score**s with 2 * fun + learning and choose.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Choose independently

Week 23 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A recommender gives 3 points for topic match and 1 for difficulty match. Book A matches topic only; B matches difficulty only. Compute **score**s and justify the **rank**ing.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Publish the features, **weight**s, raw **score**s, and reason for the top result.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Score=3×genreMatch+2×short+illustrated. Compute A=[1,0,1], B=[0,1,1], C=[1,1,0]. A reader now requires illustrations. Which book wins after applying that **constraint**?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Publish the recommender’s constraints and score explanation

> Make the preference rule and human choice inspectable.

This week’s concrete contribution

For the fictional recommender publish hard requirements, validated feature meanings, weights and the tie policy. Apply eligibility before selecting by score; preserve unknown-required-field review and no-eligible-option outcomes. Show each score component and why alternatives were excluded. A hand-specified weighted score is not a trained model, a measured learning benefit or an objective preference for every reader.

Publish the recommender’s **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion.

Keep the checkpoint, system and decision

THIS WEEK’S CHECKPOINT Publish the features, **weight**s, raw **score**s, and reason for the top result.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or **score** → **rank**ed candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

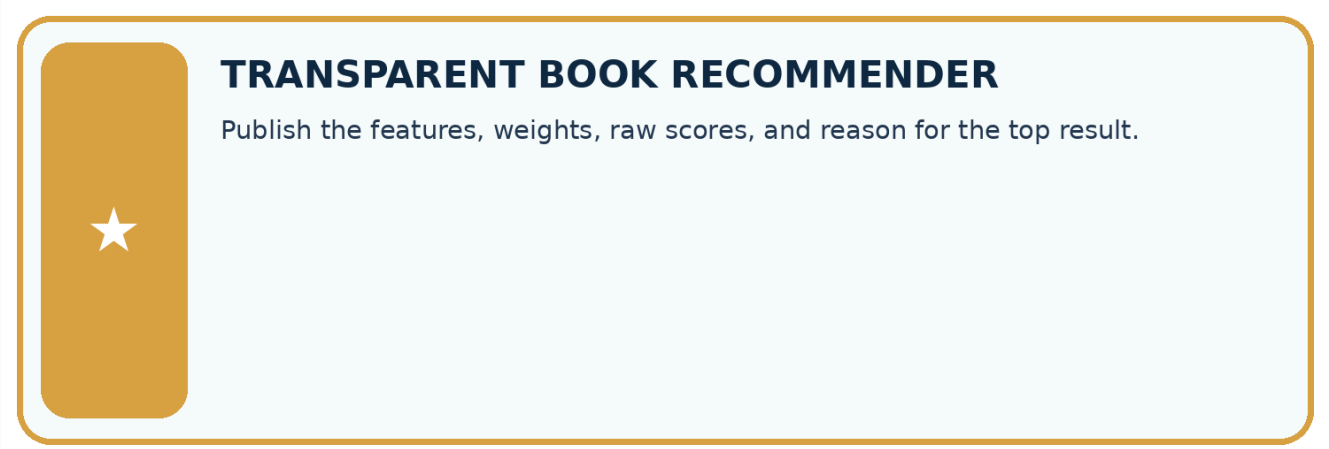
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse a **rank**ing with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: explain eligibility, score and chosen preference separately

> Retain the rejected high score without allowing it to override the rule.

Explain without notes

- Close your notes. Explain this goal in your own words: Compute a **weight**ed **score**

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
weight	A multiplier controlling how strongly a feature or value contributes to a calculation.	
score	A numerical value assigned using a specified rule; it is not automatically a probability.	
rank	Place candidates in an order using a declared comparison rule.	
tradeoff	A situation where improving one desired property can make another harder to achieve.	

Terms for the connected explanation

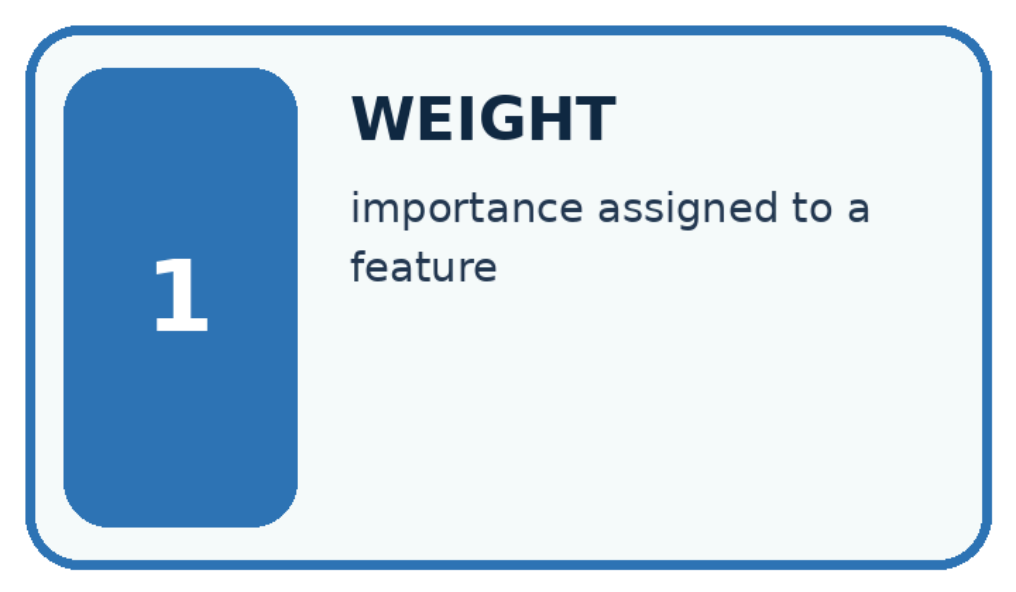
Word	Meaning
Constraint	A requirement an option must meet

▼ **Earlier wording, alternatives and picture transcripts**

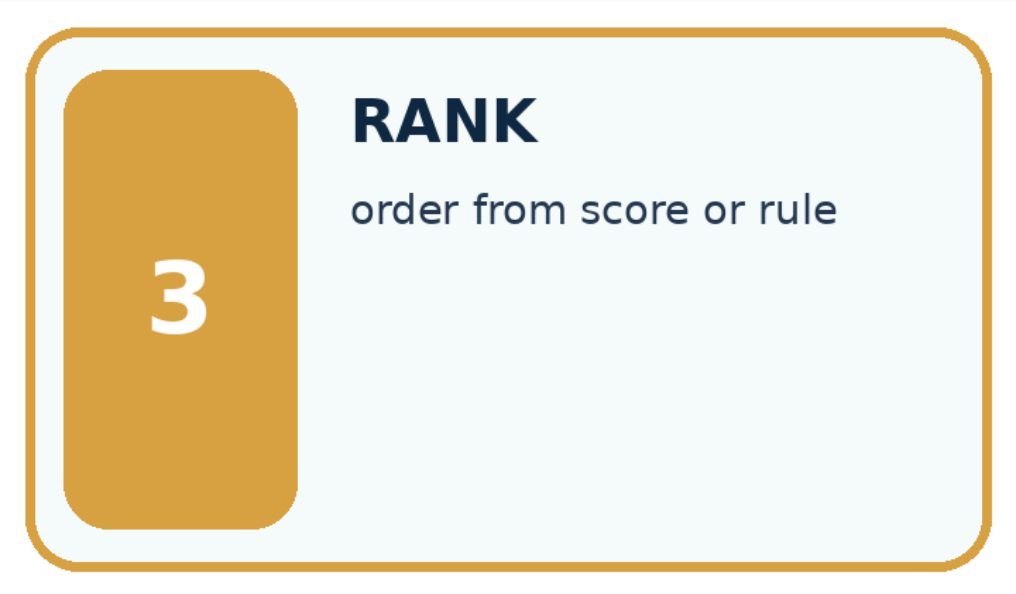
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.









Words used in the closer explanation

▼ Choose a stronger thinking challenge

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

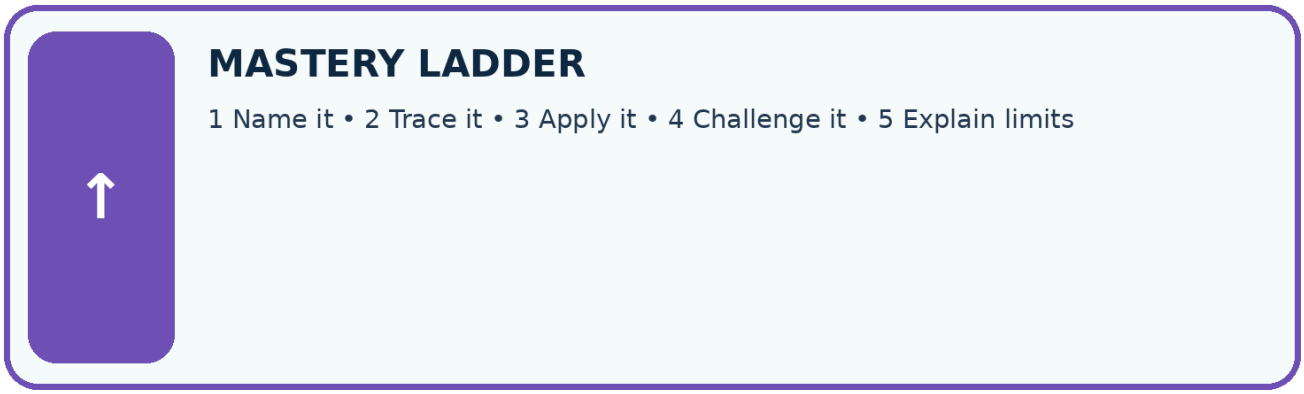
> **Write or draw your reasoning and evidence.**

▼ **Earlier wording, alternatives and picture transcripts**

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Choose independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Self-check and record

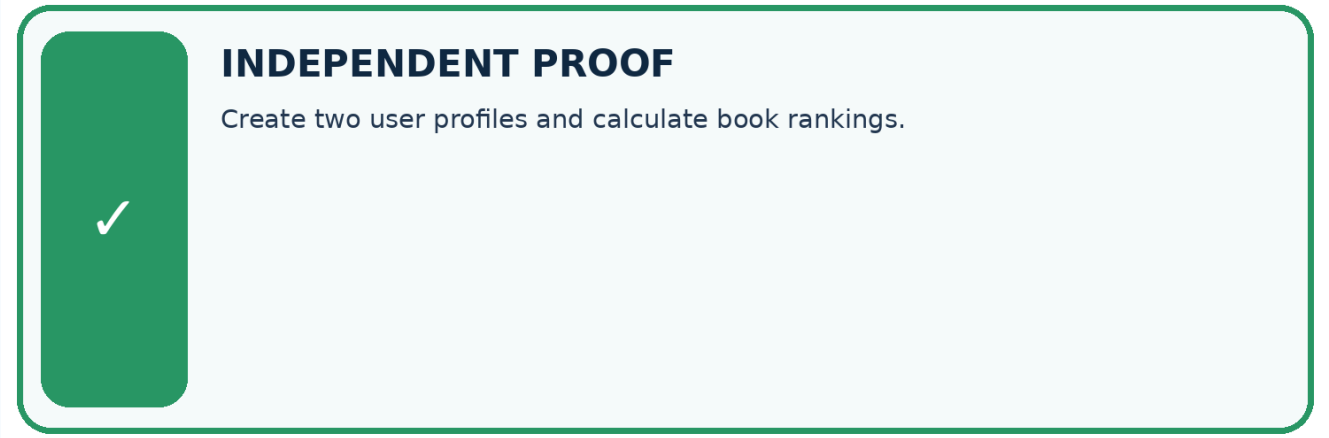
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Choose independently is the selected independent case.



EXPLANATION CHECK Apply hard requirements before **rank**ing, retain weighted components and exclusion reasons, and disclose ties and empty outcomes.

Write or draw your reasoning and evidence.

▼ **Extra practice: Weighted preferences determine the score**

REFERENCE · SECTION 13

Extra practice: Weighted preferences determine the score

> Try the question before opening its worked explanation.

Read the supplied case and question

Score=2 times genre match + length match. A matches both, B genre only, C length only.

Score and **rank**.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

A=3, B=2, C=1. **Weight**s encode a chosen preference, not universal reader truth.

Evidence or stage	Value or result
A	3
B	2
C	1

▼ Earlier wording, alternatives and picture transcripts

Week 23 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Weighted preferences determine the score

Score=2 times genre match + length match. A matches both, B genre only, C length only.

Score and **rank**.

A=3, B=2, C=1. **Weight**s encode a chosen preference, not universal reader truth.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Compute a weighted score	Explain the deciding step, show a trace or test, and state one limit.
Apply hard constraints before ranking	Explain the deciding step, show a trace or test, and state one limit.
Explain how weight choices change outcomes	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional book-activity chooser has a 14-minute limit.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional book-activity chooser has a 14-minute limit.
- Which labelled step matches this detail? B's **score** 11 cannot override its failed time requirement.
- 1. Choose
- 2. Explain exclusion
- 3. Eligibility first
- 4. **Score**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Explain exclusion B's **score** 11 cannot override its failed time requirement.

A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

Changed-case question

- Changed-case practice: Change interest **weight** from 2 to 3, keeping time limit 14. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. A **score**s 11; C also **score**s 11; tie-by-ID chooses A.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: A **score**s 11; C also **score**s 11; tie-by-ID chooses A. B remains too long; changing preferences changes only **rank**ing of A and C.

Repair and save your work

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and **rank training** neighbours. Short practice: Why do k=1 and k=3 disagree for the original query?. Return to this week and explain what you changed.

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?.

Return to this week and explain what you changed.

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

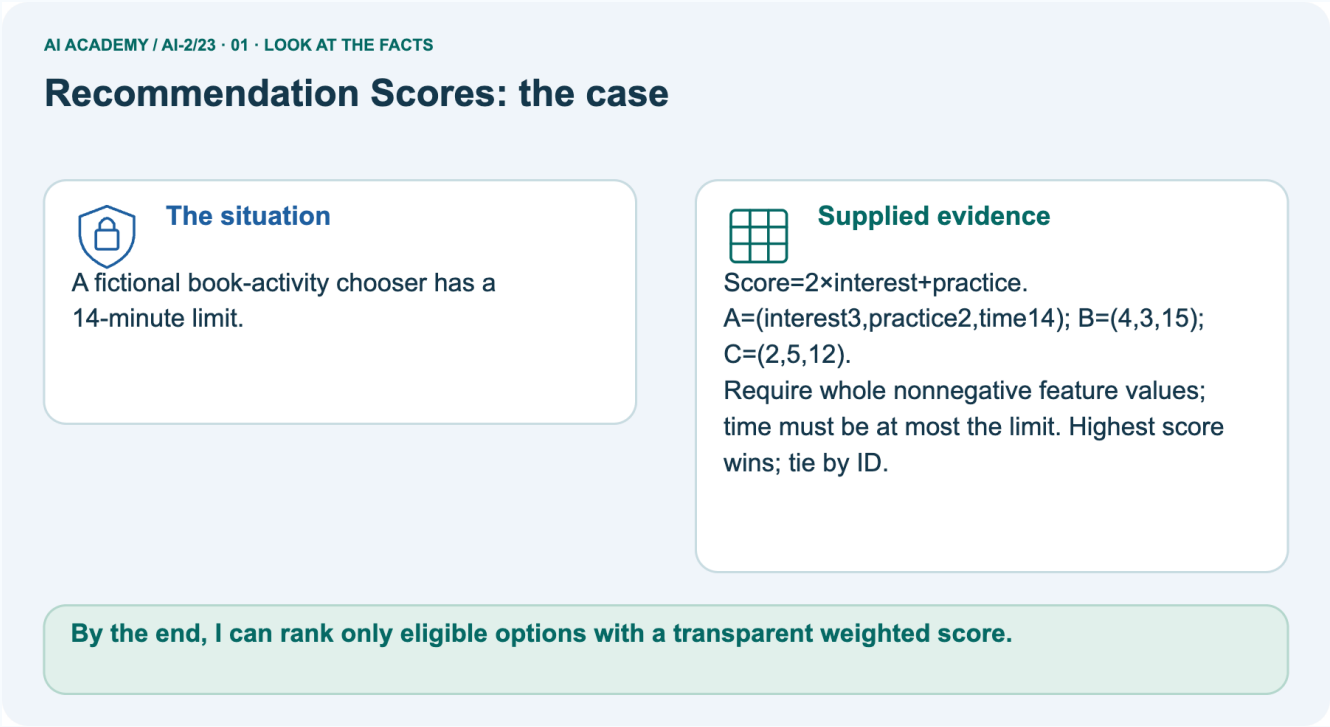
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Recommendation Scores: the case



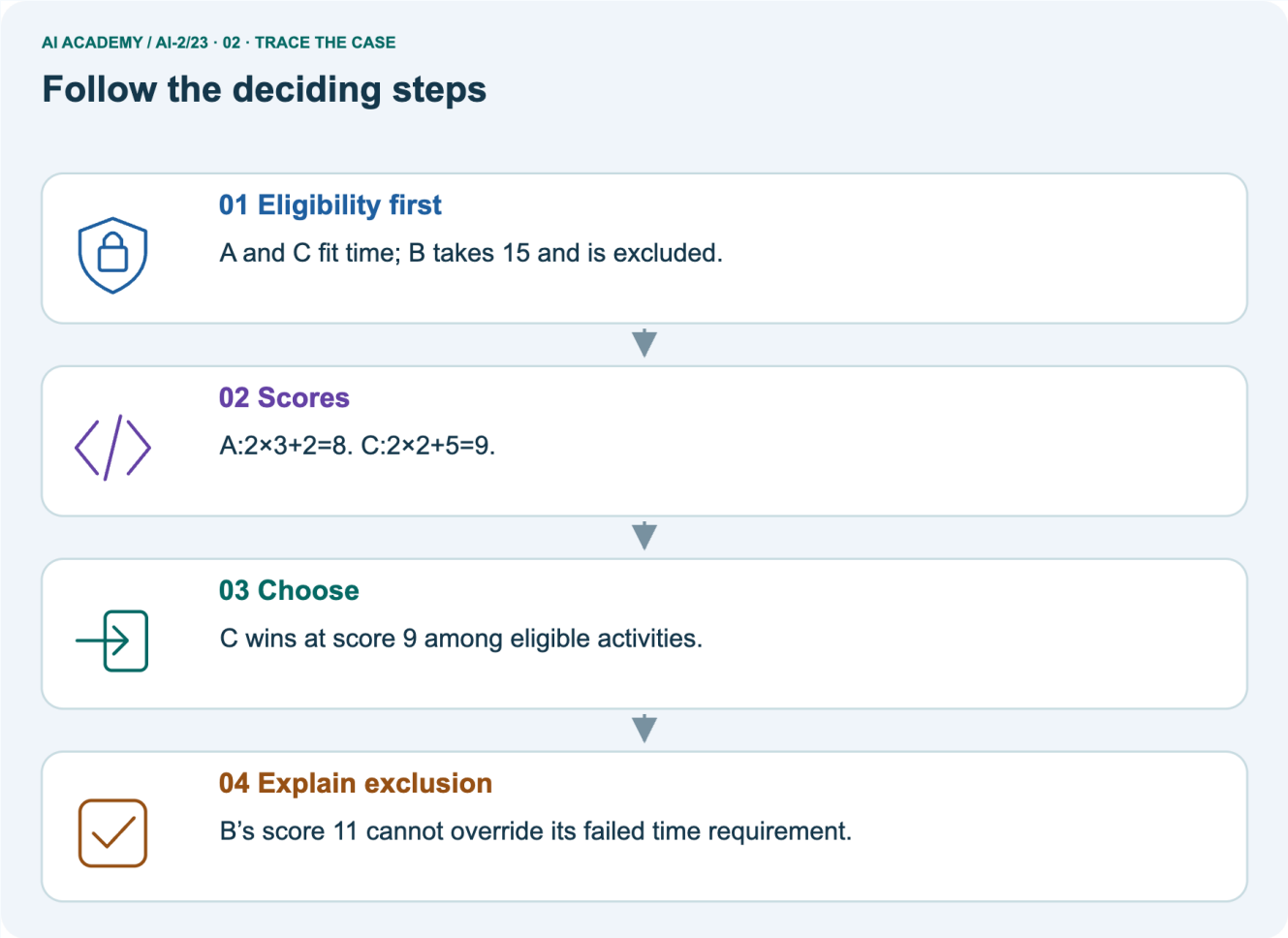
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book-activity chooser has a 14-minute limit. Supplied facts: **Score**=2×interest+practice. A=(interest3,practice2,time14); B=(4,3,15); C=(2,5,12).

Require whole nonnegative feature values; time must be at most the limit. Highest **score** wins; tie by ID.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



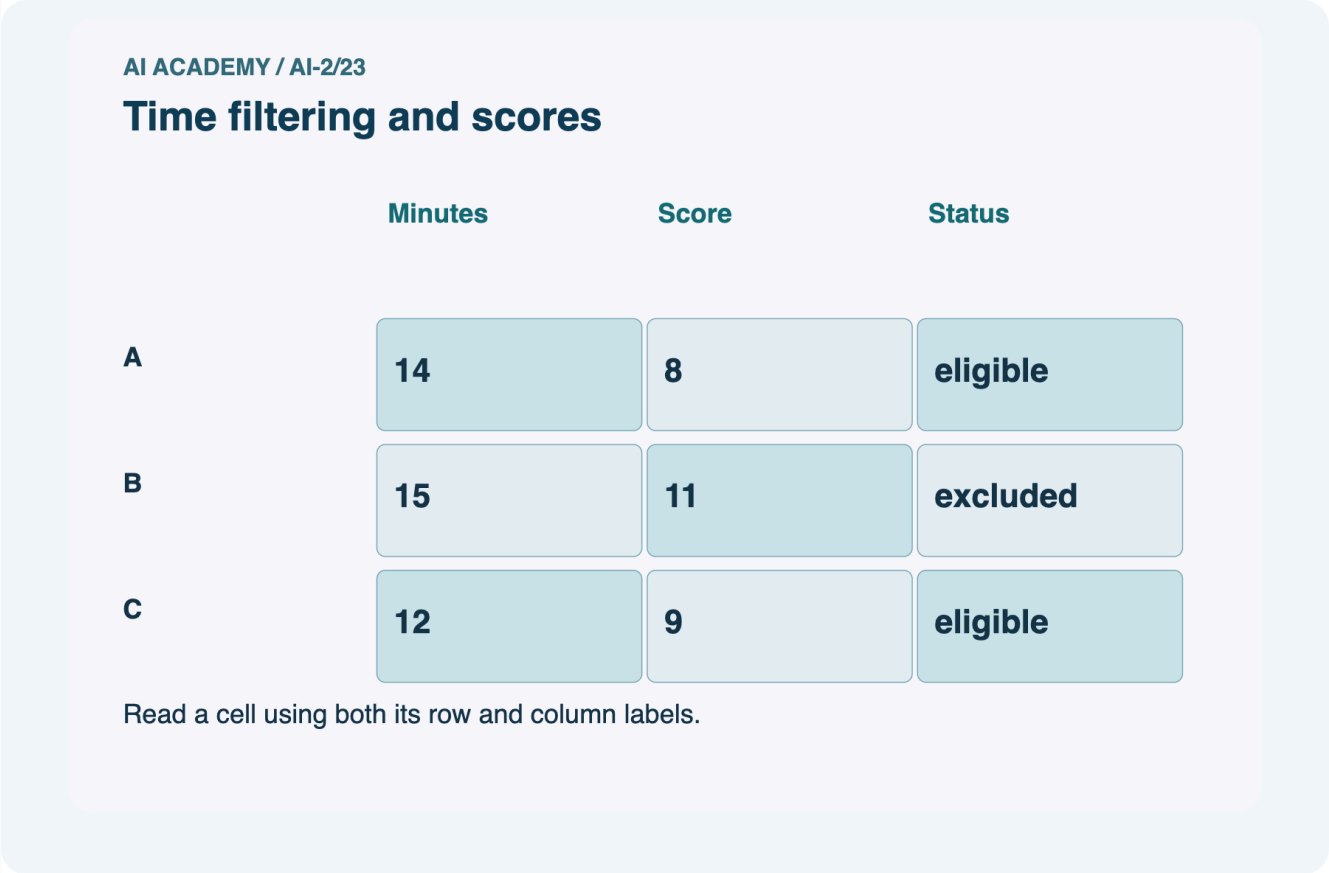
Follow the labelled picture with the matching explanation. Open the vector diagram

Eligibility first: A and C fit time; B takes 15 and is excluded. → **Score**s: A:2×3+2=8.

C:2×2+5=9. → Choose: C wins at **score** 9 among eligible activities. → Explain exclusion: B's **score** 11 cannot override its failed time requirement.

Point to each numbered stage and explain which recorded **input** it uses.

03. Time filtering and scores

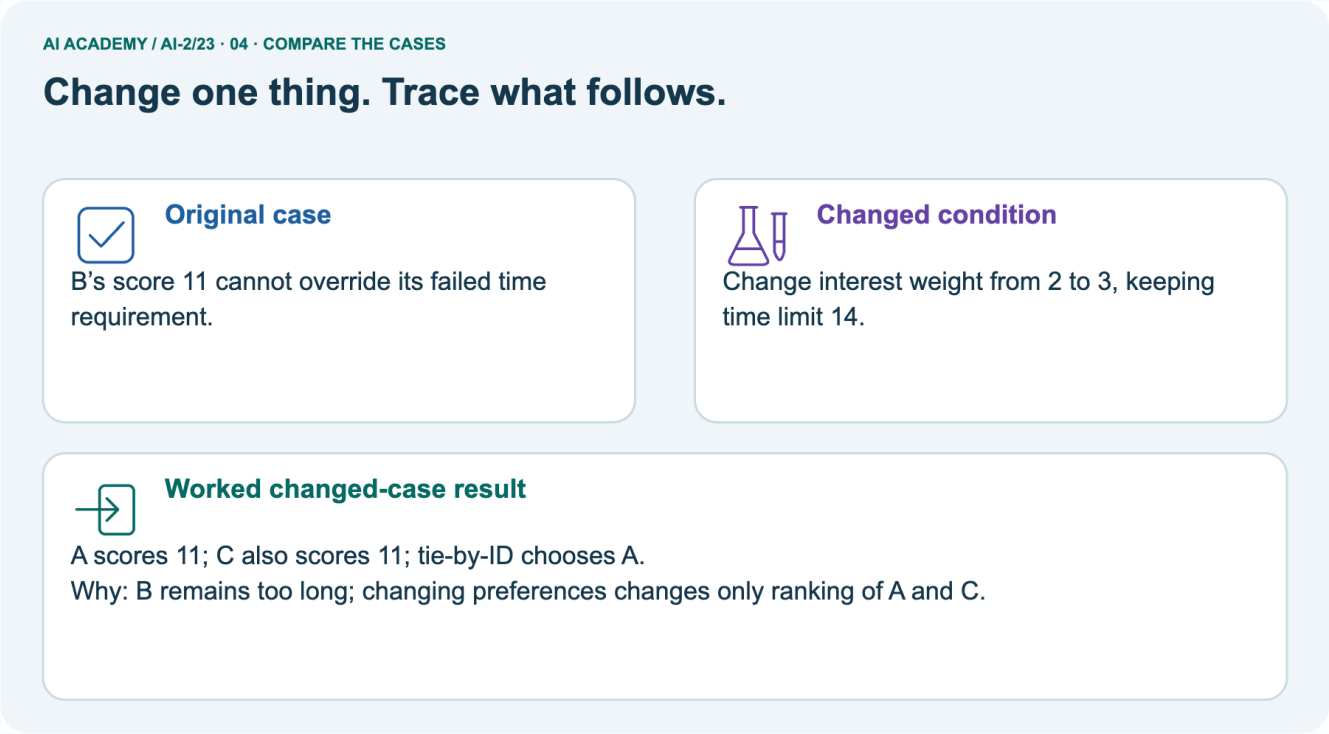


Follow the labelled picture with the matching explanation. Open the vector diagram

Time filtering and **score**s. A: Minutes = 14, **Score** = 8, Status = eligible. B: Minutes = 15, **Score** = 11, Status = excluded. C: Minutes = 12, **Score** = 9, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



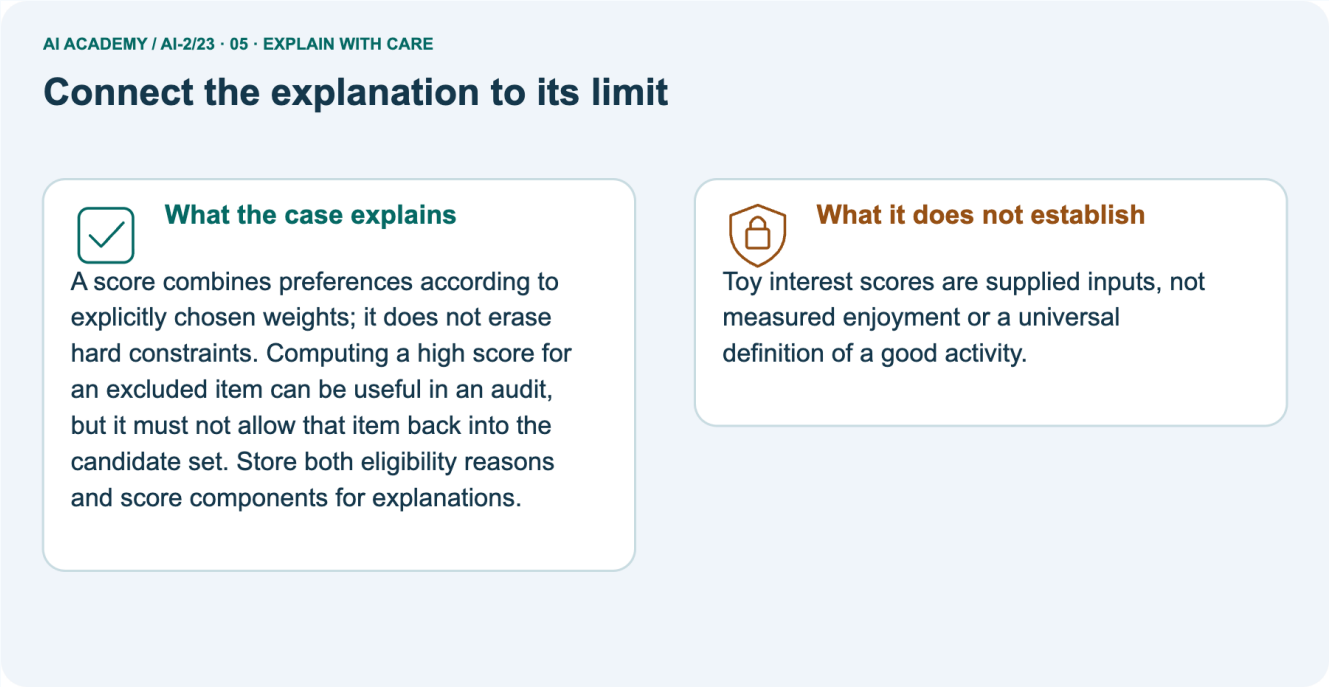
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B's **score** 11 cannot override its failed time requirement. Changed condition: Change interest **weight** from 2 to 3, keeping time limit 14. Expected result: A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

Reason: B remains too long; changing preferences changes only **rank**ing of A and C.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

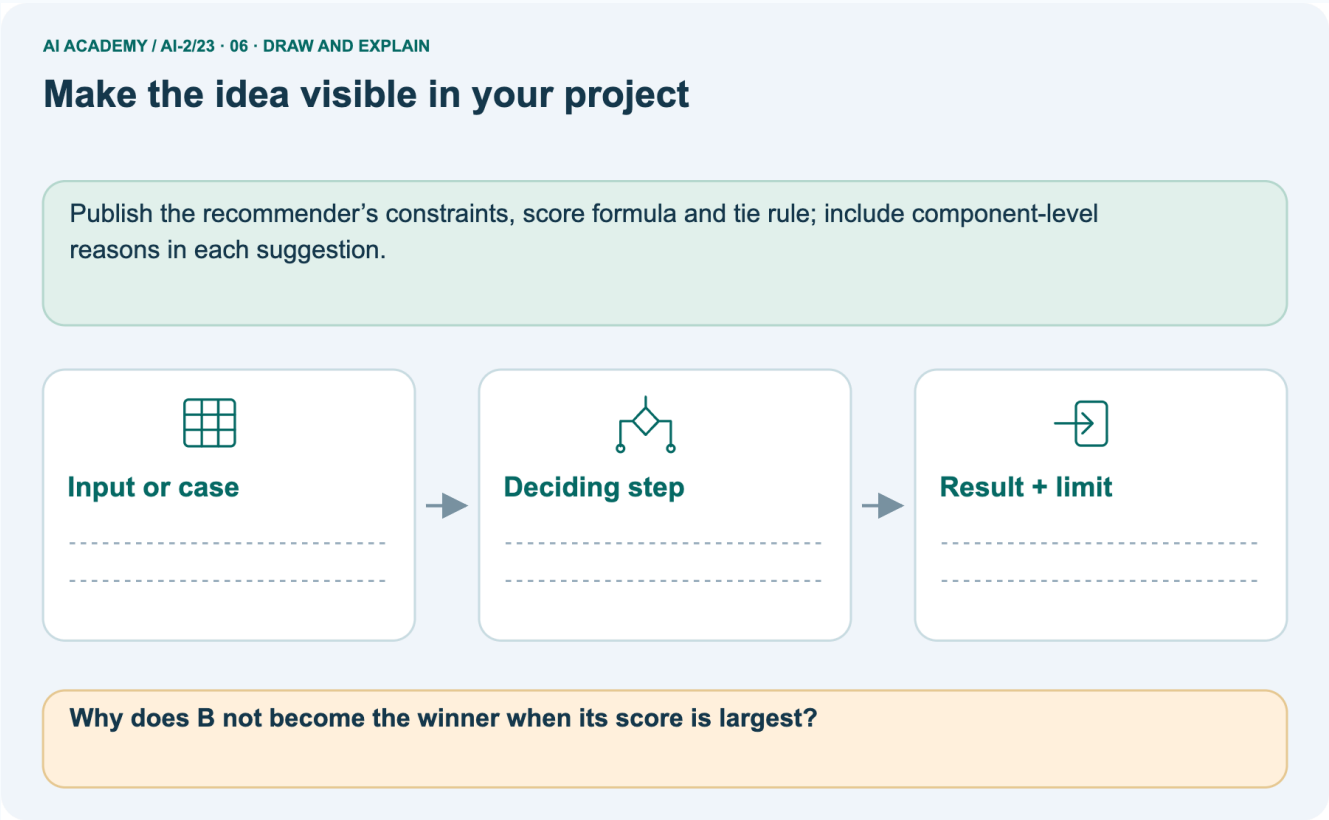
A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

Limit: Toy interest **score**s are supplied **input**s, not measured enjoyment or a universal definition of a good activity.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion. Draw your **input**, deciding step, result and limit.

Supported question: Why does B not become the winner when its **score** is largest?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Recommendation Scores: the case

02. Follow the deciding steps

03. Time filtering and scores

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.